Project Group Toolkit

An organized set of handouts and templates to guide your summer software engineering teams through collaboration, project management, and quality assurance.

Table of Contents

1. Quick-Reference
2. Fillable Checklists & Rubrics
3. Conflict-Resolution & Communication Templates
4. Roles & Responsibilities Matrix
5. Project-Management Templates
6. As-You-Go Documentation Guide

---

1. Quick-Reference

1.1 Branching & Git Workflow Quick-Reference

1. Create a branch from `main`:

```
git checkout main
git pull
git checkout -b feature/short-descriptive-name
```

2. Work & Commit Frequently:

```
git add <files>
git commit -m "Clear, concise message"
```

3. Prepare to Merge:

```
git fetch origin
git checkout main
git pull
git checkout feature/your-branch
git merge main
# resolve conflicts, run full test suite
```

4. Merge & Clean Up:

```
git checkout main
git merge --no-ff feature/your-branch
git push origin main
git branch -d feature/your-branch
```

5. Naming & Messages:

- **Branch:** `feature/login-page`, `bugfix/db-connection`
- **Commit:** "Implement password validation"

---

2. Fillable Checklists & Rubrics

2.1 Weekly Health Check

☐ Pull latest `main` and merge into your working branch
☐ Run full test suite and fix any failures
☐ Scan for TODO/FIXME comments
☐ Identify one small area for refactoring
☐ Update any outdated documentation/comments

2.2 Code Review Checklist

☐ All unit tests pass
☐ No critical compiler warnings
☐ Code adheres to established coding standards (indentation, naming conventions, comments)
☐ Documentation updated if necessary
☐ Merge conflicts resolved

2.3 Iteration Feedback Rubric

☐ No OCP or DIP violations
☐ SRP adherence confirmed
☐ No Android imports outside presentation package
☐ No database imports in presentation layer
☐ No java.sql imports outside persistence layer
☐ Encapsulation maintained (no unnecessary getters/setters)

---

3. Conflict-Resolution & Communication Templates

3.1 Stand-up Update Card

- Yesterday, I completed: _____
- Today, I will work on: _____
- Currently, I am blocked by: _____

3.2 Merge-conflict Resolution Flowchart

1. Identify conflict early
2. Assign responsible person

3. Resolve conflicts locally
4. Test thoroughly
5. Merge changes

3.3 Team Meeting Agenda Template

- Updates & Progress
- Current Roadblocks
- Key Decisions Needed
- Action Items

---

4. Roles & Responsibilities Matrix

Role	Responsibilities	Assigned To	Backup
Lead Developer	Architecture decisions, merge approvals		
Tester	Unit/integration testing, bug reports		
Documentation Lead	Maintains README, documents design decisions		
Release Manager	Version tagging, releases, publishing		

---

5. Project-Management Templates

5.1 Kanban Board (Columns)

- Backlog
- This Week
- In Progress
- Review
- Done

5.2 6-Week Gantt Chart Template

- Weekly milestones for iterations
- Deadlines for testing and integration

---

6. As-You-Go Documentation Guide

6.1 Decision-Log Template

```
Date: [YYYY-MM-DD]
Decision: [Short description]
Context:
Options Considered:
Chosen Option:
Reason:
Next Steps:
```

6.2 API Documentation Stub

```
/**
 * Brief description of the class or method.
 * @param parameter description
 * @return description of returned value
 */
```
